

Progressive Education Society's
MODERN COLLEGE OF ENGINEERING, Pune -05.
DEPARTMENT OF COMPUTER ENGINEERING

Third Year (2019 Course)

310244: Computer Networks and Security
[Semester- V]

Unit – I
Introduction To Computer Networks

Course In-charge: Ms. Pooja M. Hande

Teaching Scheme:	Credit: 03	Examination Scheme:
TH: 03 Hours/Week		Mid-Sem (TH) : 30 Marks End-Sem (TH): 70 Marks

Prerequisites Courses: --

Companion Course: Computer Networks and Security Laboratory (310247)

Course Objectives:

- To understand the fundamental concepts of networking standards, protocols and technologies
- To learn different techniques for framing, error control, flow control and routing
- To learn different layer protocols in the protocol stacks
- To understand modern network architectures with respect to design and performance
- To learn the fundamental concepts of Information Security

Course Outcomes:

On completion of the course, learners should be able to

CO1: Summarize fundamental concepts of Computer Networks, architectures, protocols and technologies

CO2: Illustrate the working and functions of data link layer

CO3: Analyze the working of different routing protocols and mechanisms

Reference & Text Books

Learning Resources
Text Books : 1. Fourauzan B.,"Data Communications and Networking", 5th Edition, Tata McGraw-Hill, Publications, ISBN:0–07 – 058408 – 7 2. Andrew S. Tanenbaum, Computer Networks, 5th Edition, Pearson India, 2012.

Course Contents

Unit I	**Introduction To Computer Networks**	**06 Hours**

Definition, **Types of Networks:** Local area networks (LAN), Metropolitan area networks (MAN),Wide area networks (WAN), Wireless networks, Networks Software, Protocol, Design issues for the Network layers.
Network Models: The OSI Reference Model, TCP/IP Model, Network Topologies, Types of Transmission Medium. **Network Architectures:** Client-Server, Peer To Peer, Hybrid.
Network Devices: Bridge, Switch, Router, Gateway, Access Point.
Line Coding Schemes: Manchester and Differential Manchester Encodings, Frequency Hopping (FHSS) and Direct Sequence Spread Spectrum (DSSS).

Introduction of Network

🗣 **A network** is a group of devices connected to share information and data. 🗣 why we need a network? Answer is very simple, to communicate with each other (share resources).

🗣 **Example-** Social networking is a group of connected people. Similar way, computer

networking is group of connected computer.

- **Node/host-** A host can be a computer, a laptop, a mobile, a internet printer or any other device that uses IP address to go to the internet.
  - **IP address-** An IP address is a numeric address. All addresses on a network require at least one unique IP address. There are two versions of IP addresses: IPv4 and IPv6.
  - **Protocols**-Protocols are rules that standardize and manage data transmission between devices. Two devices will exchange data only when they speak the same protocols.

Introduction of Network

- **Networking Devices :** The devices with main purpose of providing network facilities. Examples are Router, Switch, Access Point and Hub.

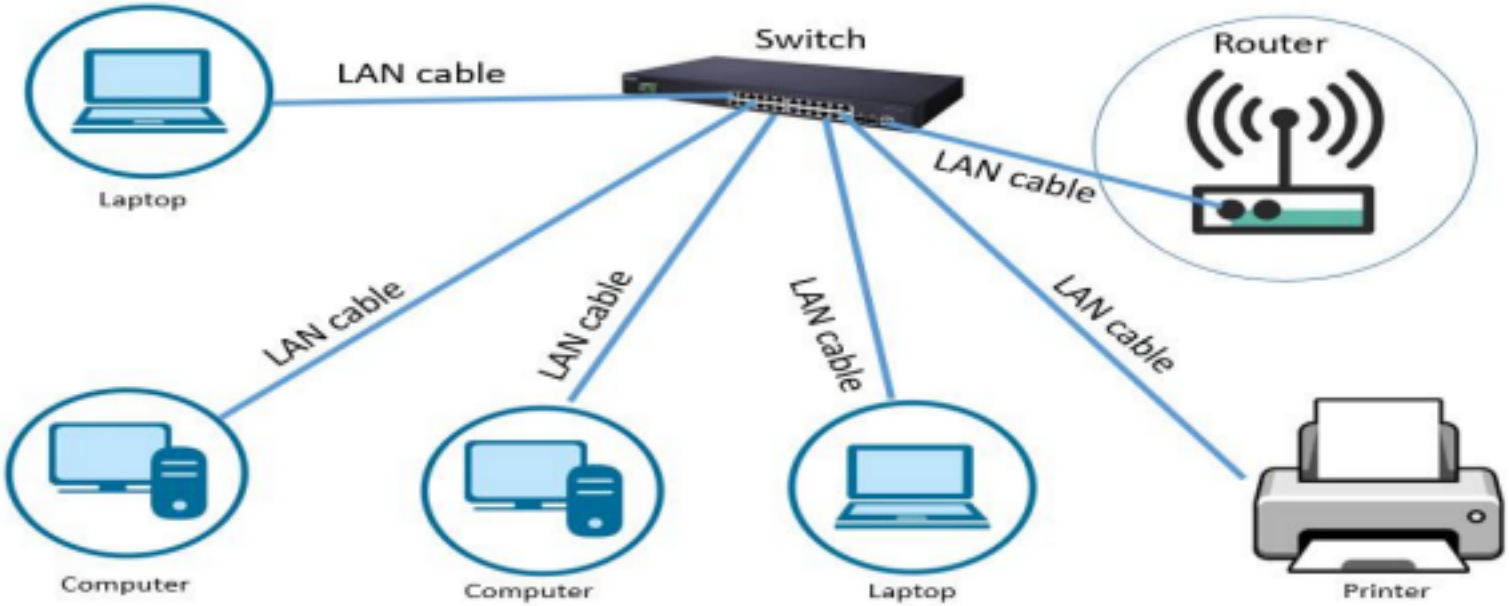
- **Networking Applications** : Software that provide an interface to communicate and share. Examples are browser, email clients and file sharing applications.
- **Types of Networks:**
  - LAN(Local Area Network)
  - MAN(Metropolitan area network)
  - WAN(Wide Area Network)
  - PAN(Personal Area Network)
  - Ad-hoc Network

LAN(Local Area Network)

● A Local Area Network (LAN) is a computer network that connects devices within a limited geographical area, such as a home, office, or campus. ● LANs enable devices to communicate with each other and share resources like printers and internet access within that confined space.

- LANs are typically privately owned and managed.
- A LAN comprises cables, access points, switches, routers, and other components that enable devices to connect to internal servers, web servers, and other LANs via wide area networks.
- **For example,** in an office with multiple departments, such as accounting, IT support, and administration, each department's computers could be logically connected to the same switch but segmented to behave as if they are separate.

LAN(Local Area Network)



Local Area Network

Advantages of LAN

- High-speed data sharing and communication
- Centralized data management
- Resource sharing (e.g., printers, storage)
- Cost-effective for small areas
- Enhances security within a closed network

Disadvantages of LAN

- Limited to a small geographic area
- Initial setup costs (hardware, wiring)
- Requires network management and maintenance
- Security risks if not properly protected

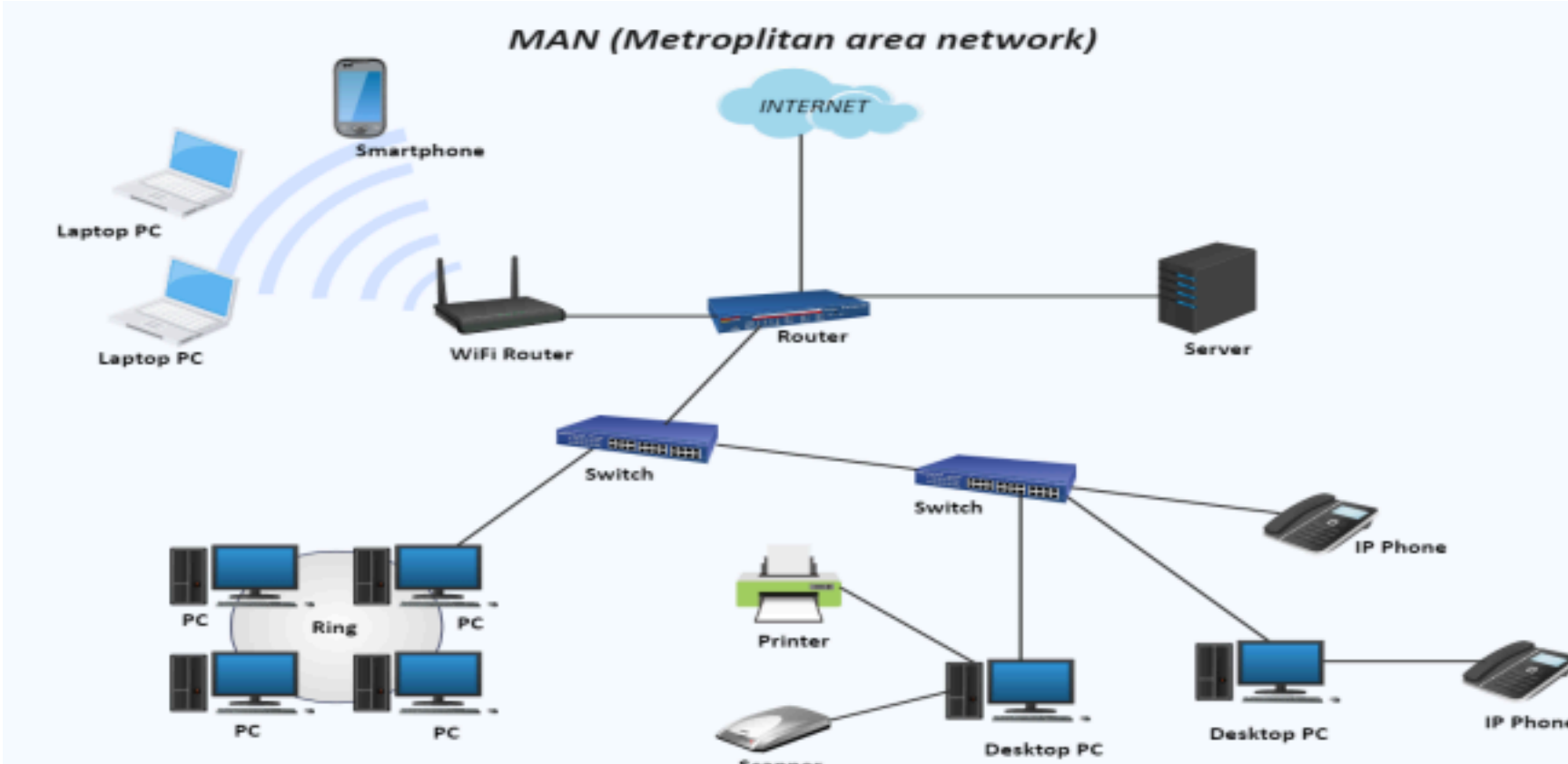
MAN(Metropolitan Area Network)

- A Metropolitan Area Network (MAN) is a computer network that connects computers within a metropolitan area, such as a city or a large campus
- . It typically spans a larger geographical area than a Local Area Network (LAN) but a

smaller area than a Wide Area Network (WAN).

- MANs are designed to provide high-speed connectivity for data, voice, and video communications within the metropolitan area.
- The size usually ranges from 5 kilometers to 50 km.
- A metropolitan area network (MAN) usually covers an entire city center or campus in computer networking. It connects LANs using high-speed backbone technologies, such as fiber optics and provides up-link services to the wide-area network (WAN) and the internet.
- Used in Digital cable television ,used in government agencies, University campuses. Cable broadband.

MAN(Metropolitan Area Network)


MAN (Metroplitan area network)
INTERNET
Smartphone
Laptop PC
Laptop PC
WiFi Router
Router
Server
Switch
Switch
IP Phone
PC
Ring
PC
PC
PC
Printer
Desktop PC
Desktop PC

Advantages of MAN

- **High Speed:** Data transfer rates for files and databases are fast due to MAN's fiber optic communication. Transmission speeds can reach 1000 Mbps quickly. ● **Internet sharing:** The installation of MANs allows users to share their internet access so that many users can have access to high-speed internet.
- **High Security:** Compared to WANs and LANs, it offers high security, the data in MAN are usually safe.
- **Less expensive:** MAN can be easily connected to the WAN network because it is less expensive. It provides strong data efficiency. All MAN data can be easily managed centrally.

Disadvantages of MAN

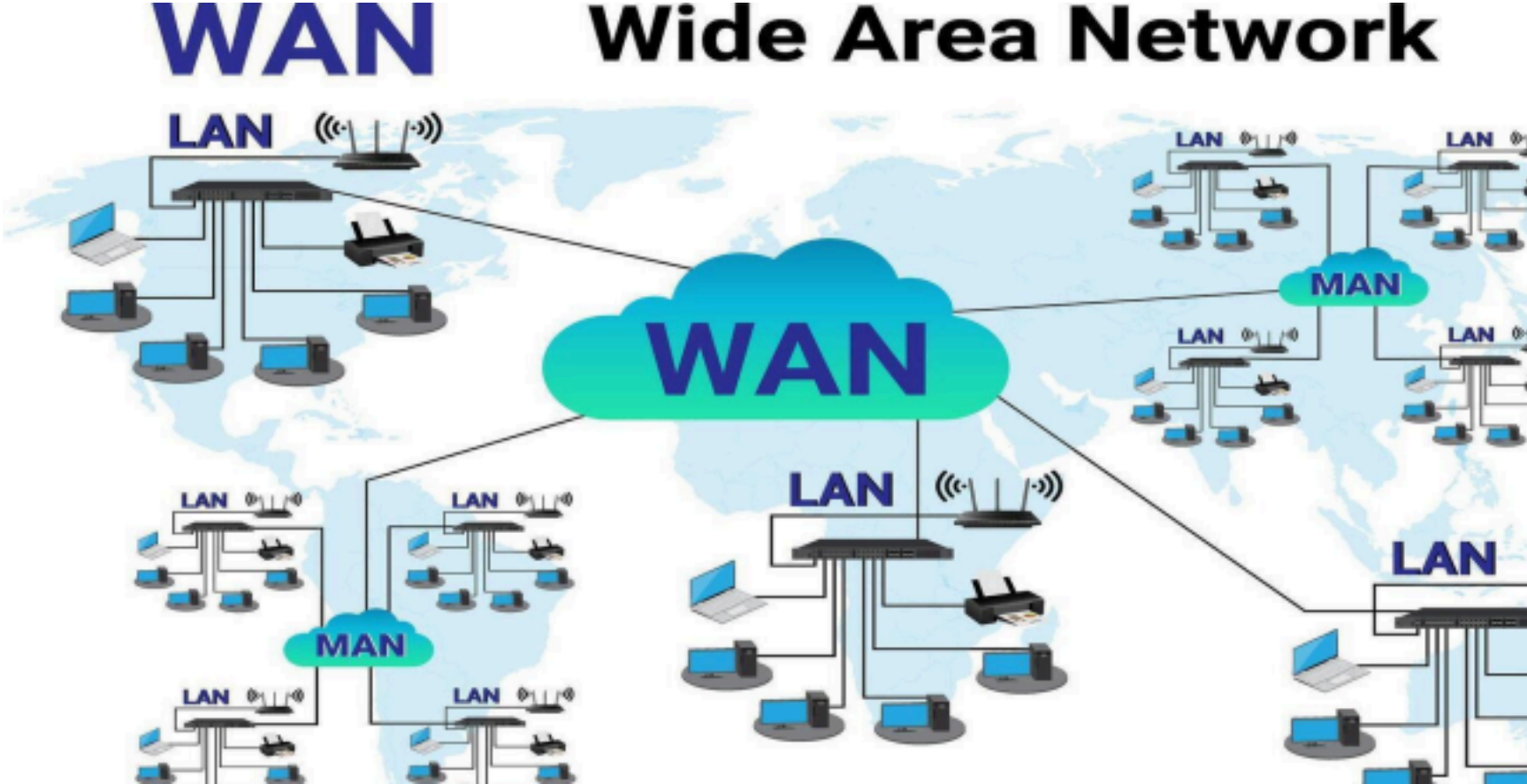
- **Internet Speed Difference:** The fiber optic network cannot use copper wires because copper wires slow down MAN. Therefore, fiber optic networks cost higher than copper wires.
- **Difficult To Manage:** As the size and number of LAN networks increase, it becomes very difficult to manage. The reason is due to additional configuration problems and security concerns.

- **High risk of hacking:** There are more risks of attacks from hackers on MAN than on LAN. Therefore, the need for high-security personnel is important.
- **Technical Staff Requires to Set up:** To configure MAN, highly technical people will be required. The technical people will be network administrators and troubleshooters.

WAN (Wide Area Network)

- The first known WAN was created by the U.S. Air Force in the late 1950s to interconnect sites in the Semi-Automatic Ground Environment (SAGE) radar defense system. An enormous network of dedicated phone lines, telephones, and modems linked the sites together.

- Any large network that spreads out over a wide geographic area is a WAN. The Internet itself is considered a WAN.
- wide area network (WAN) is a large computer network that connects groups of computers over large distances. WANs are often used by large businesses to connect their office networks; each office typically has its own local area network, or LAN, and these LANs connect via a WAN. These long connections may be formed in several different ways, including leased lines, VPNs, or IP tunnels .
- In an enterprise, a WAN is created to connect branch offices with one another or to connect remote employees working at home with the company's main office. [15]


WAN
Wide Area Network
LAN
WAN
MAN
LAN
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LAN
MAN

Advantages of WAN

- Covers a large geographical area so long distance businesses can connect on the one network.
- Shares software and resources with connecting workstations.
- Messages can be sent very quickly to anyone else on the network
- Expensive things (such as printers or phone lines to the internet) can be shared by all the computers on the network without having to buy a different peripheral for each computer.
- Everyone on the network can use the same data. This avoids problems where some users may have older information than others.

Disadvantages of WAN

- Need a good firewall to restrict outsiders from entering and disrupting the network
- Setting up a network can be an expensive, slow and complicated. The bigger the network the more expensive it is. Once set up, maintaining a network is a full-time job which requires network supervisors and technicians to be employed.

- Security is a real issue when many different people have the ability to use information from other computers.
- Protection against hackers and viruses adds more complexity and expense.

PAN-Personal Area Network

- PAN stands for personal area network, a network covering a very small area, usually a small room.
- PAN's can be wired or wireless.

🗣 The best known wireless PAN network technology is **Bluetooth**, and the most popular wired PAN is **USB**. 🗣 Wi-Fi also serves as a PAN technology, since Wi-Fi isalso used over a small area.

Personal Area Network



PAN-Advantages

- The pan is a personal network of one or two person so there is no risk of any leak of data.

PAN-Disadvantages

- The network it can only travel straight up to 10mts and if in different rooms then only 2mts.

- In the case of infrared the infra red sensor bust be in a straight line otherwise it won't communicate.
- Transmission speed is slow to moderate.

Ad-hoc Networks

- A **Wireless Ad Hoc Network (WANET)** or **Mobile Ad Hoc Network**

(MANET) is a **decentralized** type of wireless network. Unlike traditional wireless networks that rely on fixed infrastructure such as routers and access points, ad hoc networks do **not require any pre-existing infrastructure**. Instead, **each node (device) in the network acts both as a host and as a router**, forwarding data for other nodes.

- **Ad-hoc means without base station or Access point**

  (means without infrastructure)

Ad-hoc Network Diagram

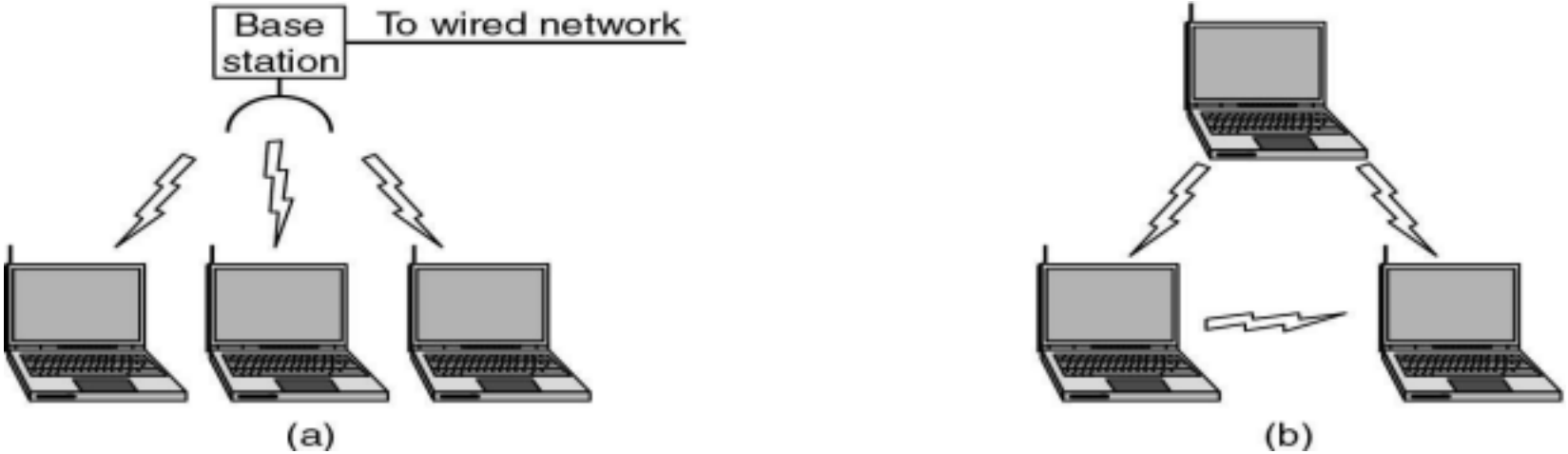


Figure (a) Wireless networking with a base station.

(b) Ad hocnetworking.

Ad-hoc Network Applications

- Group of people with laptops and they want to exchange files and data without having an access point.
- Incase if we need to exchange information and the network's infrastructure has been destroyed.
- It is suitable for military communications at battlefield where there is no network infrastructure.









Ad-hoc Network Advantages

- Ad-hoc networks can have more flexibility.
- It is better in mobility.
- It can be turn up and turn down in a very short time.
- It considered a robust network because of its non hierarchical distributed control and management mechanisms

Ad-hoc Network Disadvantages

- There are no known boundaries for the maximum range that nodes will be able to receive network frames.
- The wireless channel is weak, unreliable, and unprotected from outside interferences.
- Limited wireless range.
- Hidden terminals.
- Packet losses.

- Routes changes.
- Devices heterogeneity.

Network Devices

- Hub
- Switch
- Router
- Bridge
- Gateway
- Repeater
- Access point

- NIC
- Firewall

1. Hub



- A hub is a multiport device, which has multiple ports in a device and shares the data to multiple ports altogether. A hub acts as a dumb switch that does not know, which data needs to be forwarded where so it broadcasts or sends the data to each port.
- Hubs operate by receiving data signals from

connected devices and rebroadcasting them to all other devices. When a device sends data, the hub receives the signal and immediately broadcasts it to

approach is known as "store-and-forward" transmission, where the hub doesn't perform any data filtering or processing.[29]

Note: In the hub, data is sent to all ports but each port accepts only that data whose destination address matches their MAC address.

all other connected devices. This broadcast

Hub

- Hub is a LAN device. It works on layer 1 (physical layer). Hub is used in LAN network to connect host together. Now hubs are completely obsolete from the market.

- There are two main reasons behind this, one is hub has less number of ports. As, the network size is increasing day by day, we need to connect more devices in a LAN, so we need a device that has large number of ports.
- And the second is hubs are dummy device, simply broadcast the frame, more likely repeater.

Switch

Network switches



Switch

● Switch is a network device that connects multiple computers together in the network . ● A network switch works at the data link Layer 2 of the architecture of Open Systems Interconnection (OSI).

● These could also function where routing occurs at the network Layer 3. ● A switch has many ports to which computers are plugged in.

● When data frame arrives at any port of a network switch ,it examines the destination address. Perform some necessary checks and sends the frames to the corresponding device.

● It Support Unicast(one-to-one),Multicast(one- to-many) and Broadcast(one-to- all) communication.

- Switch can perform some error checking before forwarding data to

destination.

Switch

- Switch is a layer 2 device (data link layer). Switch is the advanced version of hub, so sometime called as intelligent hub. Switch is used to connect hosts in the LAN network.
- The basic function of a switch –to forward the frame. Switches maintain a ARP table that contains MAC address and port number.So, unlike hub switch

doesn’t broadcast every time, it receives the frame then check the table and then forward the frame. Switch broadcast the frame only when ARP table is blank.

- Now days, multi-switches are also available in the market. Multi-switch works on both layer 2 and layer 3. You can use multi-switch as a router as well as a layer 2 switch.
    - **Vendors**: Cisco, Juniper Networks, Huawei, HP, Dlink, tplink.

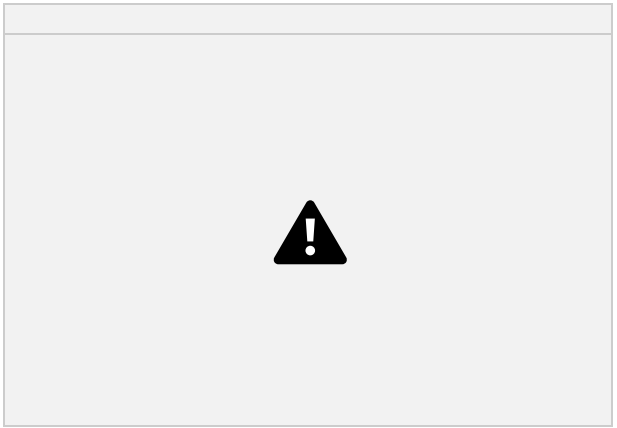
Switch

- Switch can easily identified that which device is connected with which port by using MAC Address.
- **Types of Switch**

- 1.Unmanaged Switch-inexpensive,
- 2.Managed Switch

Router

- Router is a layer 3 device (network layer). It is used to connect two different networks. The

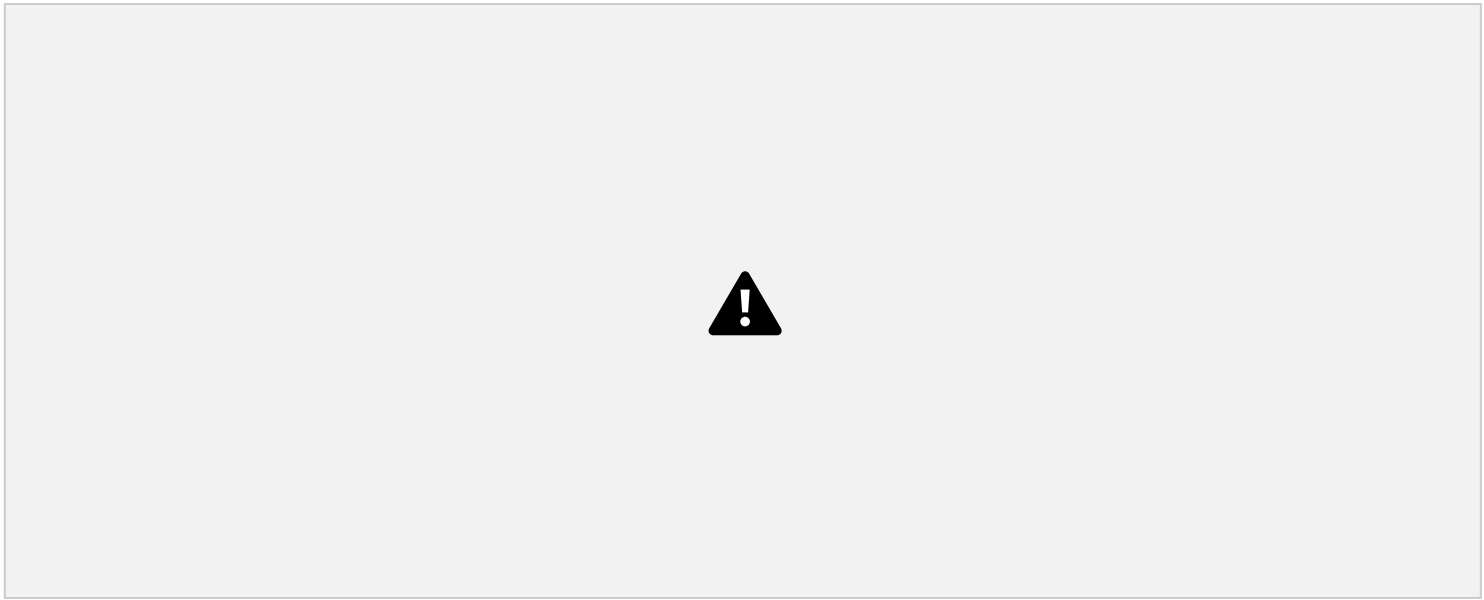


primary function of router is to forward the packets. Routers maintain a routing table that consist a list of best routes. So, whenever a packet comes, it first checks the routing table, then selects the best path and in last forwards the packet to next hop.

- **Vendors**: Cisco, Juniper Networks, Nokia.

Bridge

- Bridge is a network hardware device that used to connect the hubs together.



Bridge

- It operates at the data link layer (Layer 2) of the OSI model and is used to reduce network congestion and improve performance by managing the flow of data.
- **Network bridging** is explained as a bridge that helps to control the flow of data across segments, preventing congestion and ensuring that data is forwarded only to the correct destination.
- The primary purpose of a bridge for networking is to reduce congestion, segment the network, and improve data transfer efficiency.

Gateway

- Gateway is the entrance/exit point of a LAN network. • Gateway is basically a router. Like in our home, there is a main gate through which we enter into home or go outside to the home.
- Similar way, gateway is that router through which outside traffic comes inside and inside traffic goes outside.

Access Point



- In traditional networks, our LAN network is wired one. means all devices are connected together through wired connection. But, now the scenario is different, LAN networks have both type of hosts wired host and wireless host. To connect wireless devices to wired network, we use access points.
- In other words, we can say access points are used to merge wireless network into existing wired network

to achieve proper utilization of resources.

- Sometimes APs are used to extend the signal range (works as range-extender).
- **Wireless Link Controller(WLC)** are used to manage the Access points.

Firewall

- Firewall is a network device that is used to secure our LAN network (trusted network) from outside untrusted networks. It could be software as well as a hardware device.
- Firewall keeps an eye on the incoming and outgoing traffic and on the base of configured policies; it decides which traffic is allowed or which is denied. Firewall

sits between switch and gateway.

- **Vendors:** Paloalto, Checkpoint, Fortinet, Sophos

Modem

- Modem stands for modulation and demodulation. Modem performs both modulations as well as demodulation depends on the requirement. At receiver side, it performs demodulation, convert analog signal (electrical signal) into

binary form (100110110000011). At sender side, it performs modulation and converts digital signal into analog signal, just opposite of demodulation.

- Modulation and demodulation are opposite to each other. Here, modulation convert digital signal into analog signal and demodulation convert analog signal into digital signal.

Repeater

- Repeater is a layer 1 device. Repeater is used to achieve the original strength

of the signal. For example when we transmit a signal for a large distance, signal strength fades away as it travels more distance. So we use repeater at a fixed distance to recover the original signal strength.

- Repeater name itself defines its function, repeat whatever it receives. The only thing that you need to remember it receives weaker signal and perform some amplification and then retransmits the signal with original signal strength.

NIC(Network Interface Card)

- NIC stands for **Network Interface**

Card. It is a small circuit board (chip) on the motherboard of PC. NIC connects a computer to the internet. On NIC, there is a RJ-45 port. NIC has a MAC address (physical address) that is assigned by manufacturer.

Topology

- A network topology is the physical and logical arrangement of nodes and connections in a network , how the component of network are interconnected with each other.
- There are two types of topology
- 1.Physical Topology-(A physical network topology describes the placement of each component in the network and how they are physically connected.)
- 2. Logical Topology-(It describes how the devices in the network communicate with each other and how data is transmitted)
- Nodes usually include switches, routers and software with switch and router features. Network topologies are often represented as graphs showing node connections.

- Choosing the right topology can help increase performance, as a properly selected and maintained network topology increases energy efficiency and data transfer rates.[44]

Topology

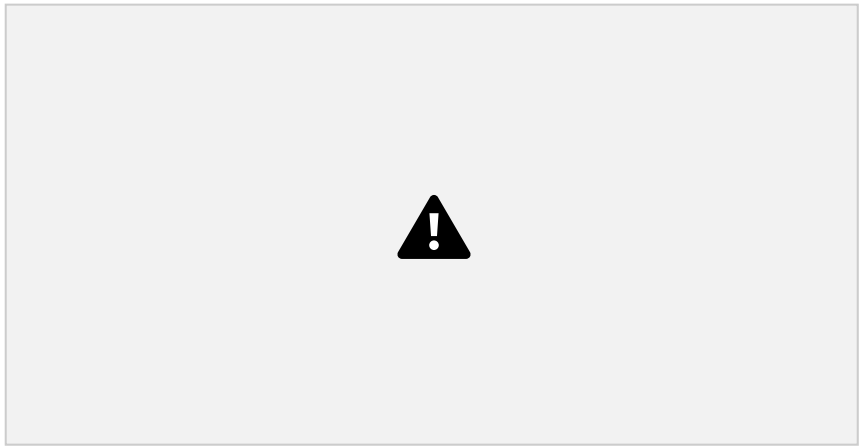


Topology



Point-to-point topology

- A point-to-point network is the simplest and most basic type of network topology. It consists of two nodes that are connected by a single link. Data travels back and forth between these two endpoints. Though easy to set up, its simplicity limits its use in modern networks.



BUS Topology

- Bus topology, alternatively known as **line topology**, is a type of network topology where all devices on a network are connected to a single cable, called a bus or backbone. This

cable serves as a shared communication line, allowing all devices (computers, printers, etc.) to receive the same signal simultaneously.

- Bus topology is **bi-directional**, i.e., data is transmitted in both directions on the backbone cable to ensure it reaches the recipient, regardless of its position on the bus. Depending on the type of network card, a twisted-pair cable (RJ-45 network cable) or a coaxial cable is utilized as a bus (backbone cable) to link network devices, computers or nodes together.

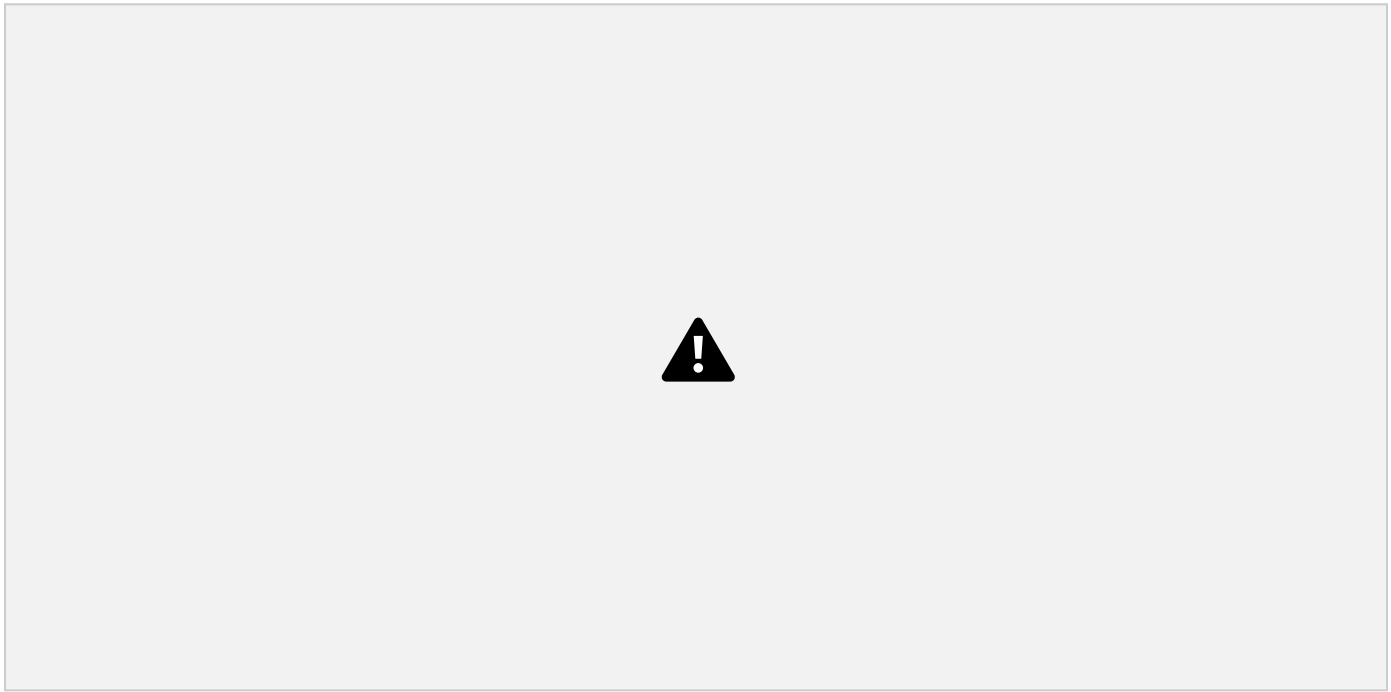
BUS Topology

How Does Bus Topology in Computer Network Work?

Bus topology has a single central cable that serves as the shared communication medium for all network devices. Each device is connected to this cable via a tap or a connector. This structure allows all devices to communicate with each other.

Terminators are used at each end of the cable to absorb signals once they reach the end of the cable. This prevents these signals' reflection down the line, which could cause interference. Here's a bus topology diagram to help you understand the concept in a better way:

BUS Topology



Bus Topology Advantages

- **Less cabling:** A single cable connects all nodes in a bus topology.
- **Easy to install:** This topology is easy to install and requires minimal configuration. ● **Less Expensive:** Bus topology is cheaper because it uses a common cable for data transmission. ● **Fast:** This topology provides high-speed data transfer as there are no intermediary devices between the nodes.
- **Upgradeable**: A new node can be added or removed in this topology without affecting the other nodes.
- **Small network:** This is best suited for situations where only a few computers are required for connection establishment.
- **Scalable:** It can be easily scaled up by adding repeaters or hubs to extend the network's range. ● **Efficient:** Bus topology is an efficient way of transmitting data, allowing multiple nodes to transmit data simultaneously.

- **Reliable:** Since all the nodes are connected to a single wire, it provides redundancy and ensures that the network remains operational even if one node fails.

Bus Topology Disadvantages

- **Core failure:** The entire network will fail if the central cable (bus or backbone) gets damaged or faulty.
- **Low security:** This is a notable security issue. All nodes in the network can hear what data is transmitted to other nodes in the network.
- **Limited speed:** The network's performance decreases as the number of devices connected increases, as all devices share the same bandwidth.
- **Not suitable for large networks:** Due to cable length limitations, bus topology is

unsuitable for large networks.

Applications of Bus Topology

- Bus topology is primarily used in small-to-medium-sized local area networks, such as small offices or home setups. It's also suitable for temporary or ad-hoc networks that require quick setup and teardown.
- In educational environments, bus topology serves as a practical model to teach basic networking concepts. Additionally, it finds applications in industrial control systems for connecting sensors, actuators, and controllers.

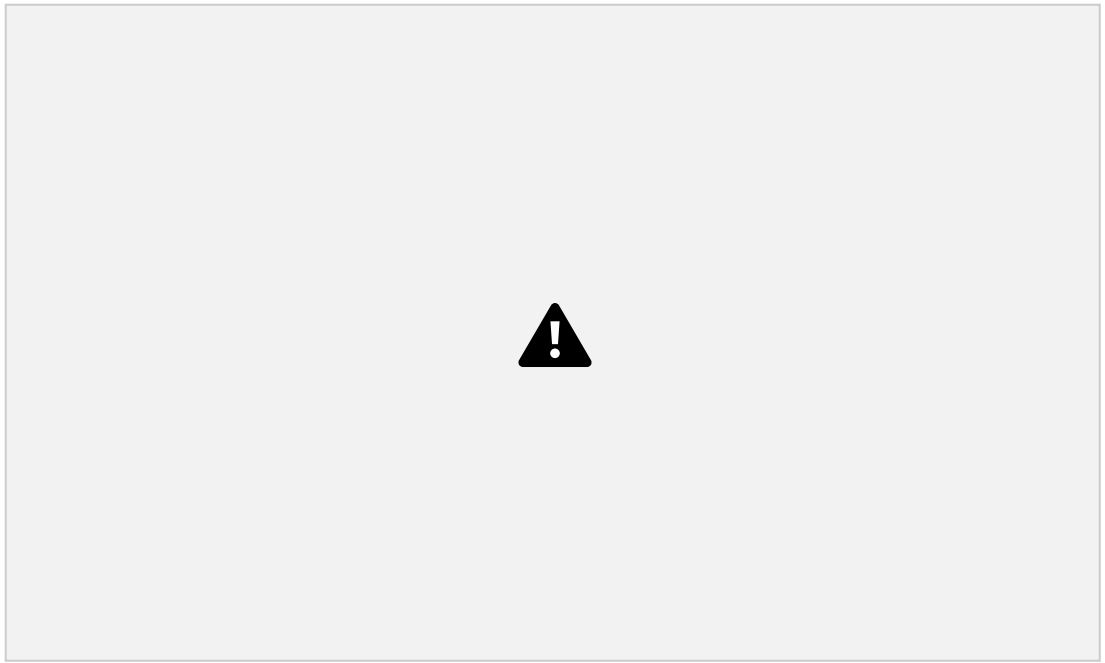
STAR Topology

- A Star topology is a type of network topology in which all the devices or nodes are physically connected to a central node such as a router, switch, or hub. The central node (hub) acts as a server, and the connecting nodes act as clients.
- It is widely used in local area networks (LANs) and other small-scale networks. ● It provides a reliable and efficient way to manage network traffic and troubleshoot

connectivity issues.

- **How Does Star Topology in Computer Network Work?**
- In Star topology, a central device is responsible for the transmission of data among nodes in the network. When data is transmitted in this topology, it is first sent to the central node (hub), which then transfers the data to all the devices on the network. Once the nodes receive the data, they check for the destination address. If the address matches, the data is processed. Otherwise, the data is discarded.

Star Topology



Star Topology Advantages

- A device failure (computer, cable) doesn't damage the entire network. Each device has its own dedicated connection, isolating issues.
- Adding or removing devices requires only a single cable connection to the central hub. There is no need to reconfigure the entire network.
- Dedicated connections prevent data packets from colliding, leading to smoother network traffic and improved performance.
- Star topology can accommodate various cable types (coaxial, fibre optic) depending on network needs and budget.

Star Topology Disadvantages

- The central hub is critical. If it fails, the entire network becomes inoperable, impacting all connected devices.
- Requires more cabling than bus topology, leading to higher installation and maintenance expenses. Central hubs can also be costly, especially for high performance models.
- Managing and maintaining the central hub requires more resources and technical

expertise than simpler topologies.

Star Topology Applications

- **Home Networks:** Uses a router as the central node, allowing devices such as computers, smart TVs, and IoT devices to connect to the internet and communicate with each other.
- **Wireless Networks:** Wireless access points (WAPs) act as the central node, with

devices connecting to the WAP wirelessly and the WAP being connected to the broader network.

- **Telecommunications:** Distributes signals from a central office to multiple users, seen in cable television networks and Internet service provider (ISP) networks.

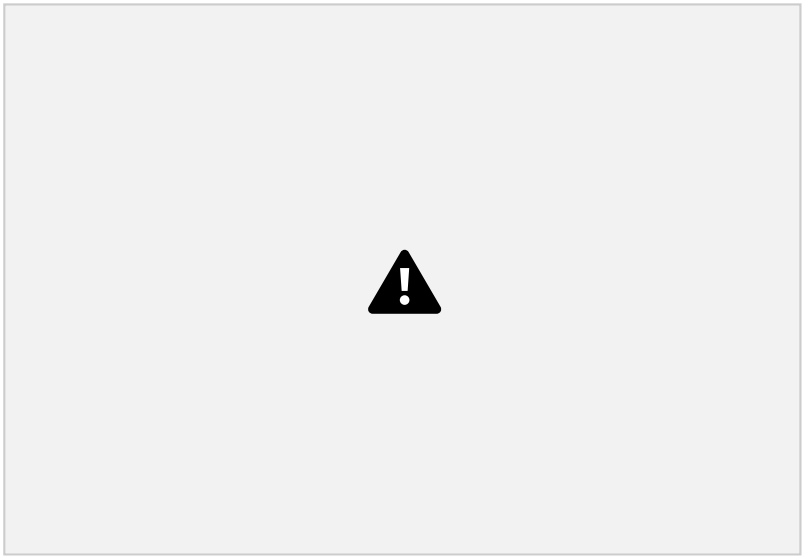
Ring Topology

- Ring topology is a closed-loop network where all devices, such as computers and printers, are connected in a circular (ring) structure.

- In this topology, each device is connected to two other devices, one on either side, forming a single continuous pathway for data transmission. The data is sent from one device to another based on their ring node's neighboring node.

● How Does Data Transmission Work in a Ring Network Topology? ● Each packet transmitted across the ring contains a destination address and the data. Each node checks the destination address as the packet circulates the ring. If the address matches the node's address, the node processes the packet. Otherwise, it passes the packet to the next node in the sequence.

Ring Topology



Note: Ring topologies can also be bi-directional, but implementing this requires additional hardware investment. However, the benefit of a bi-directional ring is that it can prevent data loss if a device in the ring fails. Whether the ring is unidirectional or bi-directional, the basic operation remains the same - sequential data transmission.

Ring Topology Advantages

- Easy to manage due to its orderly structure.
- It is cost-effective to install and expand.
- Features minimum collision, enhancing data flow efficiency.
- Speed for data transfer is notably high in this setup.
- Operates without a central server, simplifying node connectivity.

Ring Topology Disadvantages

- Troubleshooting the ring network is challenging.
- The entire network depends on a single cable's integrity.
- Adding or removing nodes can disrupt network activity and is complex.

- If one node or workstation fails, the entire network is affected.

Ring Topology Applications

- Used in both LANs and WANs for flexible data transmission.
- Utilizes bidirectional routing to maintain connectivity if a node fails. ●

Educational institutions favour it for its cost-effectiveness and reliability.

Mesh Topology

- ***Mesh topology is a type of network topology where devices are interconnected to***

multiple other devices, forming a mesh-like structure.

- Mesh topology is usually used in large institutions or enterprises as it can handle significant data traffic and be easily expanded.
- here are two types of Mesh topology:
- Full Mesh Topology
- Partial Mesh Topology

Mesh Topology

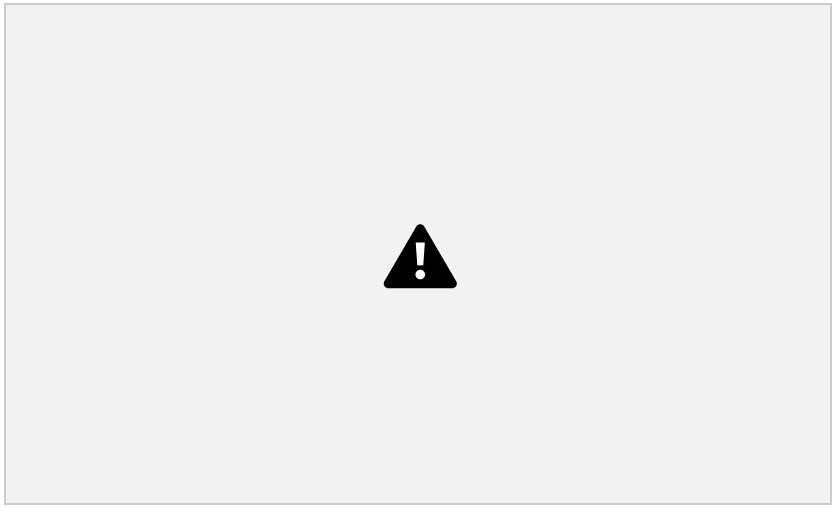
- **Full Mesh Topology:** In this type of mesh topology, every node (computer, router, switch, etc.) in the network is connected to every other node. If there are n nodes, each node will have an n-1 number of connections.

● To calculate the total number of connections, you can use this formula: n(n-1)/2. If there are ten nodes, the total number of connections will be: 10(10-1)/2 = 45. ● This design ensures that data can be routed through multiple alternative paths to reach its destination if a single connection fails.

- **Partial Mesh Topology:** In this type of mesh topology, every node in the network may not be connected to every other node.

Mesh Topology

● **How Does Data Transmission Work in Mesh Network Topology?** ● In a mesh network, data packets can take multiple routes from the source to the destination. If one path fails, the network automatically reroutes the data through an alternate path. This self-healing capability ensures the network remains operational even when the node fails.



Mesh Topology Advantages

● Even if one connection breaks, the network keeps sending data through other paths. ● The network chooses the fastest and clearest path for your data, making it run smoothly.

- Data travels directly between devices, making it harder for others to see it. - You can easily add or remove devices without messing up the whole network.

Mesh Topology Disadvantages

- Many wires and complex configurations make setting up and maintaining a mesh network harder.

- Adding devices gets expensive and complicated fast, making large networks challenging.
- Keeping all those devices running uses more power and requires more maintenance, costing you more.
- Fixing problems is trickier because there are many connections to check.
- Setting up a mesh network costs more upfront than other options.

Mesh Topology Applications

- **Smart Homes**-In smart homes, mesh networks allow devices such as lights,

thermostats, and security cameras to communicate effectively. It enhances automation and control, providing users with seamless experiences.

- **Internet of Things (IoT)**- IoT devices benefit from mesh technology as they connect and communicate over vast areas without relying on a centralized hub. It gets used in industrial settings where numerous sensors and devices deploy.
- **Public Safety Networks**-First responders can utilize mesh networks for reliable communication in emergencies. These networks adapt to changing conditions on the ground, ensuring critical information gets transmitted without interruption.

Tree Topology

- In this topology, a hierarchy is formed by connecting the root node to all other nodes. It combines characteristics of both star and bus topologies.
- In this topology, nodes are connected to a central node, which is then connected to one or more nodes at a higher level, and this pattern continues to expand outward like a tree. The topmost node is called the "root," From the root, the network branches out to secondary nodes (intermediate nodes), which may have their own connected nodes (child nodes), and so on.

● **How Does Data Transmission Work in Network Tree Topology?** ● In a network tree topology, data transmission follows a structured path. The root node serves as a central hub connected to the main server, and data moves either downwards from the root node to the leaf nodes (child nodes) or upwards from the leaf nodes towards the hierarchy,

depending on the data's destination.

Tree Topology



Tree Topology Advantages

- Easily accommodates expansion by adding new nodes or branches, facilitating growth without significant network overhaul.
- It simplifies network management and monitoring through the root node, enabling efficient updates and troubleshooting.
- It integrates various network devices and protocols, allowing for flexibility in network architecture.

Disadvantages of tree topology:

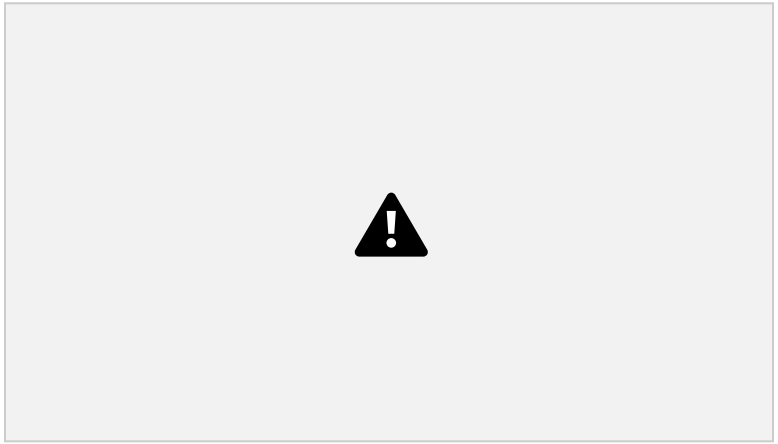
- A failure in the root node or primary branches can disrupt the entire network, leading to significant downtime.
- As the network grows, the complexity of wiring and configuration increases, requiring more maintenance effort.
- Due to the high number of nodes, the network performance of a tree topology may be

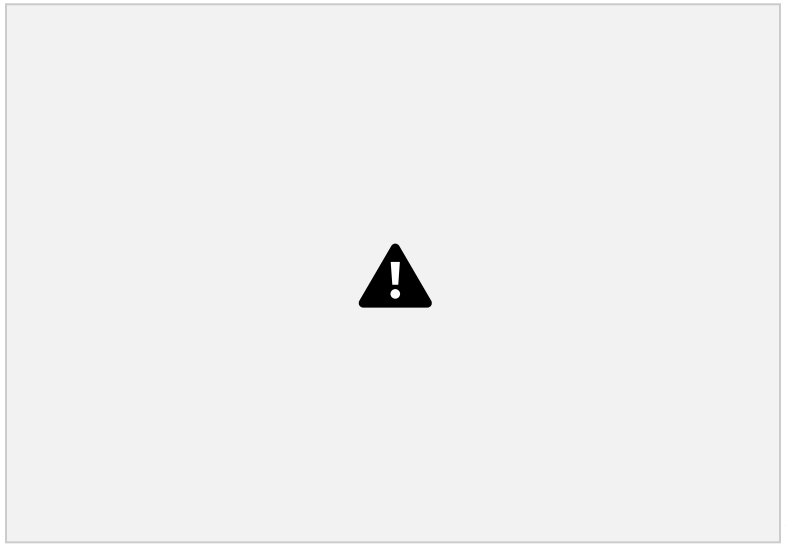
slower.

- As data passes through multiple nodes to reach distant leaf nodes, it causes traffic and communication delays.

Hybrid Topology

- A hybrid topology is a type of network topology that combines two or more network topologies, including ring, bus, and mesh topologies.





Hybrid Topology Advantages

- Adding or removing devices is easier due to the modular design of hybrid networks.

Hybrid configurations offer superior security, reliability, and scalability compared to individual topologies like star, ring, or mesh.

- Combining different topologies can increase overall network performance compared to individual topologies.

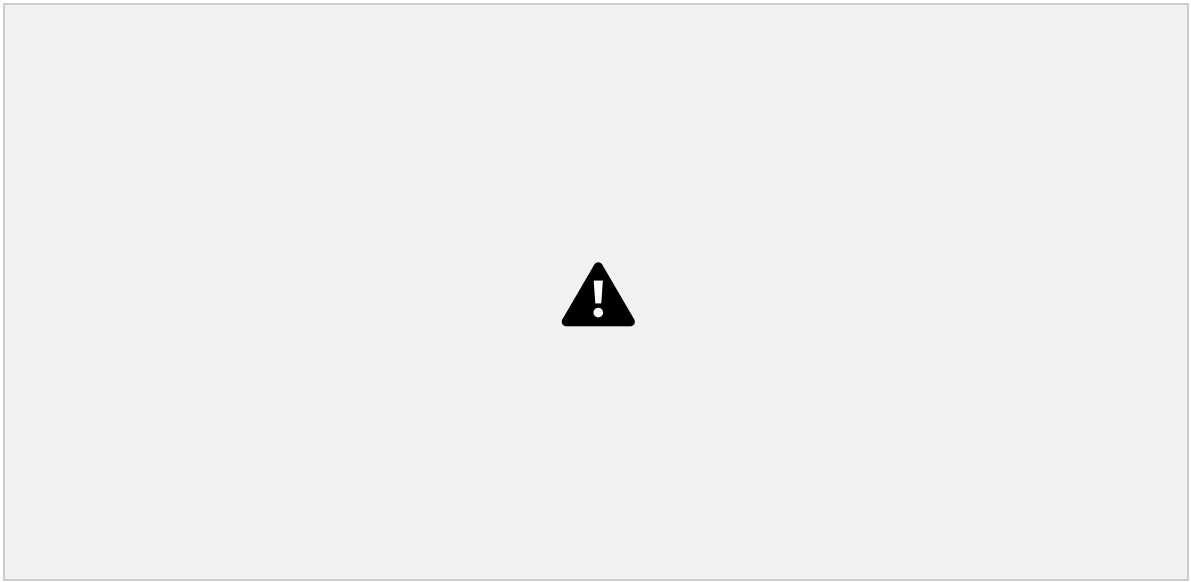
Hybrid Topology Disadvantages

- Setting up hybrid topology requires specialized hardware and bridges, which can be expensive.
- Combining different topologies requires thorough planning and expert implementation.
- Extensive cabling, additional hardware, and a larger footprint can lead to higher operational costs and potential environmental impact.

Network Architectures: Client-Server

- **Client-server architecture** is a **network model** in which two main entities — **clients** and **servers** — communicate with each other to complete specific tasks or share data.
- **Client**: The client initiates requests, waits for the server's response, and displays it to the user. For example, a web browser acts as a client when it requests a website.
- **Server**: The server processes these requests, retrieves the relevant information, and sends it back to the client. For example, a web server responds with website data when a browser requests it.

Network Architectures: Client-Server



Network Architectures: Client-Server

- **Types of Client-Server Architecture**
- **Two-Tier Architecture**: The simplest form where clients and servers communicate directly.
- **Three-Tier Architecture**: A middleware layer, often called the application layer, is added between the client and server.
- **N-Tier Architecture**: Involves multiple intermediary layers, such as security and business logic, to manage complex data processing and ensure security.

Network Architectures: Client-Server

- **Real-Life Examples of Client-Server Architecture**
- **Web Browsers and Web Servers**: Browsers like Chrome and Safari act as clients, sending requests to web servers to load pages.
- **Mobile Apps and Backend APIs**: Apps like Instagram and Amazon communicate with servers to fetch user data, posts, or product listings.
- **Gaming Platforms**: Online games use client-server models to synchronize player

actions, game state, and scores with central servers.